2. CELLPHONE CAMERA (6 points) The distance L is often called hyperfocal distance in photography and it was calculated more than one hundred years ago by Louis Derr (the figure is taken from his book *Photography for students of physics and chemistry*, published in 1906).

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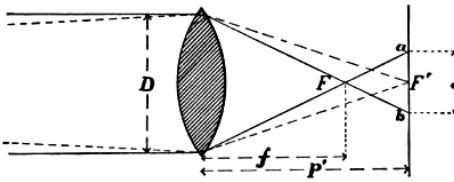


FIG. 49. Hyperfocal Distance.

Let's consider that the camera is focused to distance L and the image is formed exactly on the sensor's plane. The object's distance L and its image's distance a (corresponds to p' on the figure) are related by the lens formula $\frac{1}{L}+\frac{1}{a}=\frac{1}{f}$, thus

$$a=\frac{Lf}{L-f}=\frac{Lf}{L(1-f/L)}\approx\frac{Lf}{L}\left(1+\frac{f}{L}\right)=f+\frac{f^2}{L},$$

where the approximation $(1+x)^{-1}\approx 1-x$ (for small x) was used. Image's distance exceeds the focal length by $\Delta a=a-f=f^2/L$.

i) *(4 points)* The light coming from an infinitely far away object will pass the focal point F and form a cone which is cut by the sensor's plane. The diameter d of the cut on the sensor's plane can be found from similar triangles $d/D=\Delta a/f$, thus $d=Df/L$. Taking into account the sharpness condition $d\le\eta$, where $\eta=w/N$ is the size of a single element of the sensor, we find that the limiting value of L is $L=Df/\eta=DfN/w\approx 5.5$ m.

ii) *(2 points)* We'll now find the shortest distance s satisfying the sharpness condition. Object at distance s will have an image at distance $b=f+f^2/s$ and the light passing the lens will converge behind the sensor's plane forming a cone. The diameter d_2 of the cone's cut with the sensor's plane can be calculated from similar triangles: $d_2/D=(b-a)/b$. Accounting for sharpness condition $d_2=\eta$, we can express $b=a/(1-\eta/D)$, and substituting

the values of a and b gives

$$f + \frac{f^2}{s} = \frac{f + \frac{f^2}{L}}{1 - \eta/D} = f\,\frac{1 + \eta/D}{1 - \eta/D} \approx f\left(1 + \frac{\eta}{D}\right)^2$$
$$\approx f\left(1 + \frac{2\eta}{D}\right).$$

Finally, $f^2/s = 2f\eta/D$, or $s = \frac{1}{2}Df/\eta = L/2 \approx 2.75\,\mathrm{m}$.